

Partial fraction shortcuts

① Numerator = 1 (Just guess and divide off extra)

Example 1

$$\frac{1}{r(r+2)}$$

① Just guess a $\frac{1}{c} - \frac{1}{c}$,

$$\begin{aligned}\frac{1}{r} - \frac{1}{r+2} &= \frac{r+2-r}{r(r+2)} \\ &= \frac{2}{r(r+2)} \quad \text{--- Must be non zero constant}\end{aligned}$$

$$\therefore \frac{1}{r(r+2)} = \frac{1}{2} \left[\frac{1}{r} - \frac{1}{r+2} \right]$$

Example 2

$$\frac{1}{(ar+1)(ar+a+1)} \quad a \neq 0$$

① Just guess a $\frac{1}{c} - \frac{1}{c}$,

$$\begin{aligned}\frac{1}{(ar+1)} - \frac{1}{(ar+a+1)} &= \frac{ar+a+1 - ar - 1}{(ar+1)(ar+a+1)} \\ &= \frac{a}{(ar+1)(ar+a+1)} \quad \text{--- Must be non zero constant}\end{aligned}$$

$$\therefore \frac{1}{(ar+1)(ar+a+1)} = \frac{1}{a} \left(\frac{1}{ar+1} - \frac{1}{ar+a+1} \right)$$

② Top degree $\geq$ Bottom degree (Make top have bottom)

Example 1

$$\frac{x}{x+2}$$

$$\begin{aligned}\frac{x+2-2}{x+2} &= \frac{x+2}{x+2} - \frac{2}{x+2} \\ &= 1 - \frac{2}{x+2}\end{aligned}$$

Example 2

$$\frac{5x^2}{5x - 2}$$

① Write 2 factors (1 factor same as denominator)

Put a number to cancel off the "x" term

$(5x - 2)(x \quad)$

$-2x$

$+2x$

to cancel off the "-2x", $(5x)(\boxed{}) = 2x$

$\therefore \square = \frac{5}{2}$

$$\therefore (5x-2)(x+\frac{2}{5})$$

② Make this same as original " $5x^2$ "

$$\therefore (5x-2)\left(x+\frac{2}{5}\right) + \frac{4}{5} = 5x^2$$

$$\therefore \frac{5x^2}{5x-2} = \frac{(5x-2)(x+\frac{2}{5}) + \frac{4}{5}}{5x-2}$$

$$= (x+\frac{2}{5}) + \frac{4}{5(5x-2)} \quad \text{Ans}$$

③ Denominator factorial

Example 1

$$\frac{r+1}{(r+2)!}$$

① Make top look like bottom

$$\begin{aligned} \frac{(r+2)-1}{(r+2)!} &= \frac{(r+2)}{(r+2)!} - \frac{1}{(r+2)!} \\ &= \frac{(r+2)}{(r+2)(r+1)!} - \frac{1}{(r+2)!} \\ &= \frac{1}{(r+1)!} - \frac{1}{(r+2)!} \end{aligned}$$

④ Distinct, non-repeating factors

Example 1

$$\frac{4x+5}{(x-2)(x+1)(x-3)}$$

1) Split the fraction

$$\frac{A}{(x-2)} + \frac{B}{(x+1)} + \frac{C}{(x-3)}$$

2) To find constant over linear factor $(x-a)$, "cover up" that factor in the original func. and sub $x=a$ into whatever is left

$$\text{To find } A : x=2 \Rightarrow A = \frac{4(2)+5}{(2+1)(2-3)} = -\frac{13}{3}$$

$$\text{To find } B : x=-1 \Rightarrow B = \frac{4(-1)+5}{(-1-2)(-1-3)} = \frac{1}{12}$$

$$\text{To find } C : x=3 \Rightarrow C = \frac{4(3)+5}{(3-2)(3+1)} = \frac{17}{4}$$

$$\therefore \frac{4x+5}{(x-2)(x+1)(x-3)} = -\frac{13}{3(x-2)} + \frac{1}{12(x+1)} + \frac{17}{4(x-3)} \quad \checkmark$$

